

2nd Semester		
ISCED Code	Course Code	Course Title
0613	CSE-1221	Computer Programming 2
Credit Hours: 3	Contact Hours: 3	Type: Core, Engineering
Prerequisite: CSE-1121 (Computer Programming 1)		
Co-requisite: CSE-1222 (Computer Programming II Lab)		

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30 Marks
	SEE: Semester End Examination		50 Marks
	Total		100 Marks

Course Rationale:

Object Oriented Programming is an approach to program using Objects and Classes. This course equips the students with in-depth knowledge of different features and techniques of Object-Oriented Paradigm and hence enables them to program in Object Oriented Approach using languages like C++.

Objectives:

Upon completion of the course, students will be able to:

1. Design, compile and execute C++ programs to solve basic problems.
2. Understand basic data management in C++ programs.
3. Understand basic C++ control structures.
4. Understand Object Oriented Design.

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage (%)
1.	Demonstrate Understanding of basic OOP Concepts - Identify and explain the fundamental principles of OOP, such as abstraction, encapsulation, inheritance, and polymorphism. - Apply these concepts to design and implement simple objectoriented programs, demonstrating a clear understanding of how objects interact within a system.	10 %

2.	Apply Basic Features (Polymorphism, Inheritance) for Solving Problems - Solve programming problems using polymorphism to enable the same interface to be used for objects of different types. - Apply inheritance to create hierarchies of related classes, promoting code reuse and flexibility in design.	70 %
3.	Show Proficiency in Advanced C++ Concepts and Libraries - Demonstrate familiarity with generic functions and templates in C++, using them to write reusable and flexible code. - Utilize the Standard Template Library (STL) to work with commonly used data structures and algorithms, showcasing proficiency in its usage. - Implement exception handling mechanisms in C++ to manage runtime errors and unexpected behavior, ensuring robustness and reliability in code.	20 %

Mapping of CLO-PLO-DL-KP-EP-EA:

#	CLOs	PLOs	DL	KP	EP	EA	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
CLO1	Demonstrate Understanding of basic OOP Concepts	PLO1	C2	K3	-	-	Lecture, Class discussion, Note	Exam, Quiz, Assignment
CLO2	Apply Basic Features (Polymorphism, Inheritance) for Solving Problems.	PLO1	C3	K3	-	-	Discussion, Assignment	Assignment, Quiz, Exam
CLO3	Show Proficiency in Advanced C++ Concepts and Libraries	PLO1	C3	K3	-	-	Lecture, Assignment, Quiz	CT, Assignment

Note: DL: Domain/level of learning taxonomy, KP: Knowledge Profile, EP: Attribute of Complex Engineering Problems, EA: Attribute of Complex Engineering Activities, Learning Domains (C: Cognitive, A: Affective, P: Psychomotor)

Course Content:

#	Content	Duration	CLOs
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Mid-Term (30 Marks)			
1	Introduction: Definition of OOP, C++ Console I/O, Introduction to Classes and Objects, Basic concept of ObjectOriented Programming, Difference between Structured Programming and Object-Oriented Programming, Difference between C/C++, Benefits of OOP, Characteristics of Procedure Oriented Programming, Characteristics of Object-Oriented Programming and Applications of Object-Oriented Programming.	2 weeks	CLO1
2	Introducing Classes, Arrays, Pointers and References: Access Specifiers, Constructor and Destructor, Constructors with parameters, Object Pointers, Relation between Classes, Structures and unions, In-line functions, Automatic in-line functions, assigning objects, passing objects to functions, returning objects from function, Friend functions, Static member functions, Array of objects, Pointer to objects, this pointer, using <i>new</i> and <i>delete</i> , passing references, returning references;	2 weeks	CLO1, CLO2
3	Function Overloading: Overloading function, Constructor Overloading, Copy constructor, Default arguments, Overloading ambiguity, Address of overloaded function.	2 weeks	CLO2
Final Exam: 50 Marks Group-A (20 Marks)			
4	Operator Overloading: Binary operator overloading, Unary operator overloading, Relational and logical operator overloading, Operator overloading using friend functions, Limitations of operator overloading.	1 week	CLO2
5	Inheritance: Defining derived classes, Single inheritance, multiple inheritance, multilevel inheritance, Hierarchical inheritance, Virtual base classes, Constructors in derived classes, nesting of classes.	3 weeks	CLO2
Group-B (30 Marks)			

6	Virtual Functions: Pointers to derived classes, Applying Polymorphism using virtual functions, Polymorphic class, Pure Virtual functions, Abstract classes, early binding, and late binding.	1 week	CLO2
7	Template, Exception Handling and Standard Template Library: Generic functions, Generic classes, Exception handling, throwing mechanism, catching mechanism, Rethrowing mechanism, Specifying exceptions Templates, Components of STL, Container, Algorithms.	2 weeks	CLO3
8	C++ I/O System: Streams, Stream classes, Unformatted I/O, Binary I/O, formatted I/O, I/O manipulators, Inserters, Extractors, File I/O streams, Opening and closing files, Random access files, I/O status checking, Customized I/O and file	2 weeks	CLO3

Marks Distribution:

Course Assessment Pattern (Theory courses):

		CIE (50 marks)			SEE (50 marks)
Cognitive learning	Affective learning	Attendance Marks (10)	Assignment/ Class Test (10)	Midterm (30)	Written Exam (50)
Remember				5	5
Understand				5	5
Apply			5	10	15
Analyze				5	15
Evaluation			5	5	10
Create					
	Responding	10			
Total allocated marks		10	10	30	50

Grading Policy: As per IIUC grading policy

Learning Materials:

Text Books:

1. E. Balagurusamy, Object-Oriented Programming with C++, 4th ed., McGraw-Hill, 2008, ISBN: 0-07-066907-4.

2. D. Irvine, C++ Object-Oriented Programming, 1st ed., Prentice Hall, 1996, ISBN: 23598522.

Reference Books:

1. B. Eckel, Thinking in C++, 2nd ed., MindView, Inc., 2000.
2. R. Lafore, Object-Oriented Programming in C++, 4th ed., Sams Publishing, Indianapolis, Indiana, 2002.
3. H. Schildt, Teach Yourself C++, 2nd ed., McGraw-Hill, 2005, ISBN: 9780074638705.

2nd Semester		
ISCED Code	Course Code	Course Title
0613	CSE-1222	Computer Programming 2 Lab
Credit Hours: 1.5	Contact Hours: 3	Type: Core, Engineering
Prerequisite: CSE-1122 (Computer Programming 1 Lab)		
Co-requisite: CSE-1221 (Computer Programming 2)		

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30 Marks
	SEE: Semester End Examination		50 Marks
	Total		100 Marks

Course Rationale:

Object Oriented Programming is an approach to program using Objects and Classes. This course equips the students with in-depth knowledge of different features and techniques of Object-Oriented Paradigm and hence enables them to program in Object Oriented Approach using languages like C++.

Objectives:

- Upon completion of the course, students will be able to:
1. Design, compile and execute C++ programs to solve basic problems.
 2. Understand basic data management in C++ programs.
 3. Understand basic C++ control structures.
 4. Understand Object Oriented Design.

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

Department of Computer Science and Engineering (CSE)

#	CLO Description	Weightage (%)
1.	Demonstrate the understanding of basic OOP Concepts - Identify and explain the fundamental principles of OOP, such as abstraction, encapsulation, inheritance, and polymorphism. - Apply these concepts to design and implement simple objectoriented programs, demonstrating a clear understanding of how objects interact within a system.	10%
2.	Solve real life problems using basic concepts of OOP - Solve programming problems using polymorphism to enable the same interface to be used for objects of different types. - Apply inheritance to create hierarchies of related classes, promoting code reuse and flexibility in design.	30%
3.	Develop Application Integrating Modern Data Handling and Exception Handling techniques Develop robust applications that incorporate file handling for data manipulation and exception handling for error management, ensuring reliability and resilience in handling various scenarios.	60%

Mapping of CLO-PLO-DL-KP-EP-EA:

#	CLOs	PLOs	DL	KP	EP	EA	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
CLO1	Demonstrate Understanding of basic OOP Concepts	PO1	C3	K3	-	-	Lecture, Class discussion, Lab work, Note	Assignment, Class performance, Exam
CLO2	Solve real life problems using basic concepts of OOP	PO1	C3	K3	-	-	Lecture, Class discussion, Lab work, Note	Assignment, Class performance, Exam
CLO3	Develop Application Integrating Modern Data Handling and	PO5	C6	K6	-	-	Demo interview, Presentation	Project, Presentation, Report

	Exception Handling techniques							
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Note: **DL:** Domain/level of learning taxonomy, **KP:** Knowledge Profile, **EP:** Attribute of Complex Engineering Problems, **EA:** Attribute of Complex Engineering Activities, **Learning Domains** (C: Cognitive, A: Affective, P: Psychomotor)

Course Content:

#	Content	Duration	CLOs
1	Introduction: Definition of OOP, C++ Console I/O, Introduction to Classes and Objects, Basic concept of ObjectOriented Programming, Difference between Structured Programming and Object-Oriented Programming, Difference between C/C++, Benefits of OOP, Characteristics of Procedure Oriented Programming, Characteristics of Object-Oriented Programming and Applications of Object-Oriented Programming.	2 weeks	CLO1
2	Introducing Classes, Arrays, Pointers and References: Access Specifiers, Constructor and Destructor, Constructors with parameters, Object Pointers, Relation between Classes, Structures and unions, In-line functions, Automatic in-line functions, assigning objects, passing objects to functions, returning objects from function, Friend functions, Static member functions, Array of objects, Pointer to objects, this pointer, using <i>new</i> and <i>delete</i> , passing references, returning references;	2 weeks	CLO1, CLO2
3	Function and Constructor Overloading: Function overloading, constructor overloading, copy constructor, Default arguments, overloading ambiguity, overloaded function addresses.	2 weeks	CLO2
4	Operator Overloading: Binary/unary operator overloading, Relational and logical operator overloading using friend functions, Limitations of operator overloading.	1 week	CLO2
5	Inheritance: Single, multiple, multilevel, and hierarchical inheritance, Constructors in derived classes, class nesting.	3 weeks	CLO2
6	Virtual Functions: Pointers to derived classes, Applying Polymorphism using virtual functions, Polymorphic class, Pure Virtual functions, Abstract classes, early binding, and late binding.	1 week	CLO2

7	Template, Exception Handling and Standard Template Library: Generic functions, Generic classes, Exception handling, throwing mechanism, catching mechanism, Rethrowing mechanism, Specifying exceptions Templates, Components of STL, Container, Algorithms.	2 weeks	CLO3
8	C++ I/O System: Streams, Stream classes, Unformatted I/O, Binary I/O, formatted I/O, I/O manipulators, Inserters, Extractors, File I/O streams, Opening and closing files, Random access files, I/O status checking, Customized I/O and file	2 weeks	CLO3

Marks Distribution:

Course Assessment Pattern (Lab courses):

Bloom's Category		Evaluations out of 100 marks				
		CIE (50 marks)				SEE (50 marks)
Cognitive learning	Affective learning	Attendance Marks (10)	Assignment/ Lab Test (10)	Mid Lab (20)	External Participation in Curricular/CoCurricular Activities (10)	Lab Exam (50)
Remember						
Understand					5	5
Apply			5	10		15
Analyze					5	
Evaluation			5	10		30
Create						
	Responding	10				
Total allocated marks		10	10	20	10	50

Grading Policy: As per IIUC grading policy

Learning Materials:

Text Books:

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